



down transformer X1. This 15V secondary output of X1 is fed to bridge rectifier BR1. Capacitor C1 filters the voltage ripples, and the filtered output is supplied to the voltage regulator LM7812 (IC1). IC1 provides a 12V regulated DC output to power IC2. LED1 serves as a power-on indicator. Both timers of the dual timer LM556 (IC2A and IC2B) are configured in an astable multivibrator mode.

LED strip dimmer

The circuit of the LED strip dimmer is built around IC2A, the first timer of the LM556. IC2A is configured in an astable multivibrator mode, generating a square wave with adjustable high and low times. These times can be varied by changing resistor R2, potentiometer VR1, and capacitor C3. A 100-kilo-ohm potentiometer (VR1) is used as a variable resistor to adjust the duty cycle of the output signal. Capacitor C3, which charges through diode D1 and discharges through diode D2, produces a pulse width modulated (PWM) signal at the output pin 5 of the LM556. Consequently, MOSFET T1 turns on and



Fig. 3: LED strip used in the prototype

off, causing the LED strip to turn on and off.

The duty cycle of the PWM signal generated by C3 can be adjusted using potentiometer VR1, allowing control over the brightness of the LED lights. The 12V LED strip is connected across CON2. Fig. 3 shows the LED strip used during the circuit testing. The maximum wattage of this strip is approximately 4.8 watt per metre.

Call bell

The circuit of the call bell is built around IC2B, the second timer of the LM556. It is also wired in the astable multivibrator mode.

When the circuit is powered on by pressing S1, capacitor C5 starts charging through resistors R5 and R6. Once the voltage across C5 reaches 2/3 of Vcc, the threshold pin (pin 12) causes the discharge pin (pin 13) to go low, discharging the capacitor. When the voltage across C5 falls to 1/3 of Vcc, the trigger pin (pin 8) makes the discharge pin (pin 13) go high again, allowing the capacitor to charge again. This cycle repeats, generating a continuous square wave output on pin 9.

This square wave output at pin 9 of drive transistor T1 enables

speaker LS1 to generate an audible signal. In other words, when someone presses switch S1, the call bell is activated. When switch S1 is released, reset pin 10 of IC2B goes low, disabling IC2B. As IC2B stops producing the square wave output, no sound is produced through speaker LS1.

Construction and testing

You can assemble the circuit on a general-purpose PCB and solder

it. Thereafter, enclose it in a suitable box. Install the LED strip at a convenient location and place switch S1 of the call bell at the desired position. Position speaker LS1 near the attendant(s) using twin flexible wires. When someone presses switch S1, the call bell will sound. Connect the 12V LED strip across CON2 and place the potentiometer VR1 at the front for easy adjustment of the LED brightness.

To check the operation of the circuit, follow these steps:

Connect the mains power supply across CON1. When LED1 is lit, it indicates that the power is on and the circuit is ready for use. Adjust potentiometer VR1 to vary the brightness of the LED strip. Test the call bell by pressing switch S1 to see if speaker LS1 produces sound.

After confirming that the circuit successfully performs both the dimming and call bell functions, it is ready for use.

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/make-led-strip-dimmer-call-bell> **EFY**

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PARTS LIST

Semiconductors:

IC1	- LM7812, 12V voltage regulator
IC2	- LM556 dual timer
T1	- IRFZ44 N-channel MOSFET
T2	- BC337 NPN transistor
D1, D2	- 1N4148 signal diode
LED1	- 5mm LED
BR1	- 1A bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2, R3, R7	- 1.2-kilo-ohm
R4	- 3.3-kilo-ohm
R5	- 10-kilo-ohm
R6	- 5.6-kilo-ohm
VR1	- 100-kilo-ohm potentiometer

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2, C4	- 0.01 μ F, ceramic disk
C3, C5	- 0.1 μ F, ceramic disk

Miscellaneous:

CON1, CON2	- 2-pin connector
S1	- Push to on switch
LS1	- 8-ohm, 0.5-watt speaker
X1	- 230V AC primary to 15V, 500mA secondary transformer
	- 12V LED strip